

A positive Poisson-representation theorem with one nonscalar row

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Status. Verified partial result; likely valid subject to human review. It gives a new dimension class in which the open question of Popescu [1] has an affirmative answer.

1 The question and the result

For the regular polyball $\mathbf{B}_n$, $n = (n_1, \dots, n_k)$, Popescu asks whether every positive free k -pluriharmonic function is the Poisson transform of a completely positive map on the C^* -algebra generated by the right universal model.

An open question remains. Is any positive free k -pluriharmonic function on the regular polyball $\mathbf{B}_n$ the Poisson transform of a completely positive linear map? The answer is positive if $k = 1$ (see [23]) and also when $n_1 = \dots = n_k$ (see Section 3).

The cited Proposition 3.8 actually proves $k = 1$ and the polydisk case $n_1 = \dots = n_k = 1$. (The last displayed source sentence appears to omit the final “= 1”.) The following strictly enlarges those proved cases.

Theorem 1. *Let $n \in \mathbb{N}^k$ have at most one coordinate larger than one. Every positive free k -pluriharmonic function*

$$F : \mathbf{B}_n(\mathcal{H}) \longrightarrow B(\mathcal{E}) \otimes_{\min} B(\mathcal{H})$$

is the noncommutative Poisson transform $\mathcal{P}\mu$ of a completely positive map $\mu : C^(\mathbf{R}) \rightarrow B(\mathcal{E})$.*

Thus arbitrary operator-valued coefficients and arbitrarily many scalar coordinates are allowed, together with one genuinely free row of any finite size.

2 The simultaneous unitary-extension lemma

The only new dilation input is elementary but useful.

Lemma 1. *Let $U_1, \dots, U_s$ be commuting isometries on $\mathcal{K}$, and let $V = (V_1, \dots, V_n)$ be a row isometry on $\mathcal{K}$ such that every V_j commutes with every U_i . There are a Hilbert space $\tilde{\mathcal{K}} \supset \mathcal{K}$, commuting unitaries $\tilde{U}_1, \dots, \tilde{U}_s$, and a row isometry $\tilde{V} = (\tilde{V}_1, \dots, \tilde{V}_n)$ extending the original operators, such that the tuple $(\tilde{V}, \tilde{U}_1, \dots, \tilde{U}_s)$ is doubly commuting.*

Proof. For $p \leq q$ in $\mathbb{N}^s$, use the isometry $U^{q-p} = U_1^{q_1-p_1} \dots U_s^{q_s-p_s}$ as the connecting map from a copy $\mathcal{K}_p$ of $\mathcal{K}$ to $\mathcal{K}_q$. Let $\tilde{\mathcal{K}}$ be the Hilbert direct limit and $J_p : \mathcal{K} \rightarrow \tilde{\mathcal{K}}$ its canonical isometries, normalized by

$$J_p = J_q U^{q-p} \quad (p \leq q).$$

Set, initially on the dense union of the ranges of the J_p ,

$$\tilde{U}_i J_p h = J_p U_i h, \quad \tilde{V}_j J_p h = J_p V_j h.$$

The definitions are independent of the representative because all original operators commute. Inner products show that the $\tilde{V}_j$ are isometries with mutually orthogonal ranges. Each $\tilde{U}_i$ is an isometry, and it is onto because $\tilde{U}_i J_{p+e_i} h = J_p h$; hence it is unitary. The extended operators commute. Fuglede’s theorem now gives $\tilde{V}_j \tilde{U}_i^* = \tilde{U}_i^* \tilde{V}_j$. The commuting unitaries also commute with one another’s adjoints, so the rows are doubly commuting. $\square$

3 Proof of Theorem 1

First suppose $F(0) = I_{\mathcal{E}}$. The positive-structure theorem in Section 3 of [1] supplies a Hilbert space $\mathcal{K} \supset \mathcal{E}$ and commuting row isometries $\mathbf{V} = (V_1, \dots, V_k)$ whose compressed moments are exactly the coefficients of F :

$$A_{(\alpha;\beta)} = P_{\mathcal{E}} \mathbf{V}_{\alpha}^* \mathbf{V}_{\beta} \text{nid}_{\mathcal{E}}. \quad (1)$$

After relabeling, V_1 is the possible nonscalar row, while $V_2, \dots, V_k$ are single commuting isometries. Lemma 1 extends this tuple to a doubly commuting row-isometric tuple $\tilde{\mathbf{V}}$ on $\tilde{\mathcal{K}}$.

Because $\tilde{\mathbf{V}}$ extends $\mathbf{V}$, its analytic products agree with those of $\mathbf{V}$ on $\mathcal{K}$. Hence all compressed moments in (1) are unchanged. Double commutation makes

$$\pi(\mathbf{R}_{\alpha} \mathbf{R}_{\beta}^*) := \tilde{\mathbf{V}}_{\alpha} \tilde{\mathbf{V}}_{\beta}^*$$

a $*$ -representation of $C^*(\mathbf{R})$. If $J : \mathcal{E} \hookrightarrow \tilde{\mathcal{K}}$ is inclusion, then $\mu(T) = J^* \pi(T) J$ is completely positive. Its moments are (1), so uniqueness of the free pluriharmonic coefficients gives $F = \mathcal{P}\mu$.

For general $A := F(0) \geq 0$ and $\varepsilon > 0$, normalize

$$G_{\varepsilon} = (A + \varepsilon I)^{-1/2} (F + \varepsilon I) (A + \varepsilon I)^{-1/2}.$$

The normalized case, followed by conjugation with $(A + \varepsilon I)^{1/2}$, produces a completely positive map μ_{ε} representing $F + \varepsilon I$ and satisfying $\|\mu_{\varepsilon}\| = \|A + \varepsilon I\|$. A point-ultraweak cluster limit as $\varepsilon \downarrow 0$ is completely positive. Its moments converge to the coefficients of F , and therefore it represents F . This proves the theorem. $\square$

4 Why the full arbitrary-row case remains

The proof uses that the semigroup generated by the scalar rows is $\mathbb{N}^s$: its isometric representation has the direct-limit unitary extension above. With two nonscalar rows one would need to Cuntz-dilate one free row while extending the other commuting row and preserving its orthogonal-range relations. There is no analogous Ore direct limit, and the defect space of a row isometry need not be invariant under the other row. General product-system regular-dilation theorems impose additional positivity or regularity hypotheses. Consequently the argument does not settle the original question for two or more genuinely free rows.

A bounded search using the exact question and this dilation formulation found no later primary-source answer or this intermediate dimension theorem. The novelty claim is therefore only *apparently new within the bounded search*. Eight focused upgrade attempts, including finite polynomial counterexample searches and an attempted iterative Cuntz dilation, are recorded in the accompanying attempt file.

References

- [1] G. Popescu, *Free pluriharmonic functions on noncommutative polyballs*, Analysis & PDE **9** (2016), 1185–1234; arXiv:1511.08852.